

Magnetic resonance imaging (MRI) is a common non-invasive medical imaging method with high resolution and soft tissue contrast. The main component of an MRI scanner is a very strong superconducting magnet with the magnetic field strength measured in Teslas (T). The magnet resembles a large pipe, 1.25 m long, open on both ends, with a bore diameter of 0.6 m.

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Gas lawnmower	100	250
Vacuum cleaner	70	3,000

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$$f_c = \frac{1}{2\pi RC}$$

Equation 1

MRI can be combined with another imaging method, positron emission tomography (PET), to detect cancer. PET imaging maps the uptake of a radioactive metabolite. The radioactive decay emits a positron that annihilates with an electron and forms gamma rays that are imaged by the scanner.

Which of the following best describes the direction of the magnetic force on a positron traveling inside the magnetic field of an MRI scanner? (Note: Assume that the direction of the positron's velocity is not the same as the direction of the magnetic field.)

- ☐ A. It is in the same direction as the magnetic field. [12%]
- ✔ ☒ B. It is perpendicular to the magnetic field. [68%]
- ☐ C. It is in the same direction as the positron velocity. [8%]
- ☐ D. It is in the opposite direction as the positron velocity. [10%]

Correct

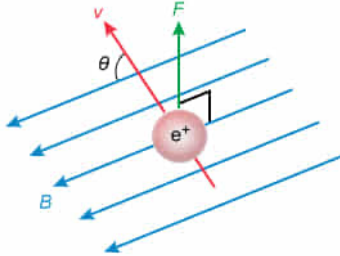
67% answered correctly

Explanation:

Force on a charge moving in a magnetic field

$$F = qvB \sin \theta$$

- F Lorentz force
- q Charge
- v Speed of charge
- B Magnetic field
- θ Angle between B and v



F is perpendicular to B and v

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The **Lorentz force** is the magnetic force that acts on a **charge** that is moving within a **magnetic field**. The Lorentz force F , velocity v of the charge, and magnetic field B are all **vectors** that have both magnitudes and directions. The magnitude of F on a charge q depends on the angle θ between the directions of v and B according to the equation:

$$F = qvB \sin \theta$$

The direction of F is **perpendicular** to both the direction of v and the direction of B . Consequently, the direction of F is determined by using the **right-hand rule**.

In this question, a positron traveling inside an MRI scanner will experience an F perpendicular to both v and B . Therefore, F must be perpendicular to B .

(Choice A) The magnetic force is perpendicular to the magnetic field, not in the same direction as the magnetic field.

(Choices C and D) The magnetic force is perpendicular to the velocity of the positron; it is not in the same direction or the opposite direction.

Educational objective:

The Lorentz force is the magnetic force exerted on a moving charge inside a magnetic field. The direction of this force is perpendicular to both the magnetic field and the velocity of the charge.

Time spent: 0 sec

Subject:
Physics

QID: 403877

System:
Electrostatics and Circuits

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Equation 1

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How many times greater is the intensity of the sound from a 1.0 T MRI scanner compared to the intensity of the sound from a vacuum cleaner?

- ☐ A. 40 [23%]
- ☐ B. 100 [12%]
- ☐ C. 1,000 [10%]

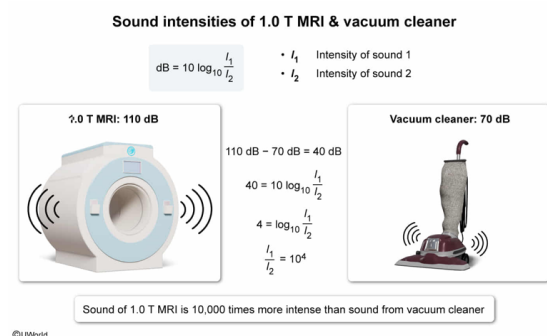


- ☒ D. 10,000 [52%]

Correct

53% answered correctly

Explanation:



The **intensity** of a sound is defined as the power of the sound wave per unit area. The perceived loudness of sound is approximately **logarithmic**, so increasing the intensity of a sound by a factor of ten is perceived as being two times louder. Sound intensities are measured on the **decibel (dB) scale** relative to the threshold of normal hearing. A sound with intensity I_1 relative to the threshold intensity I_2 is defined in dB as:

$$\text{dB} = 10 \log_{10} \frac{I_1}{I_2}$$

Therefore, I_1 has an intensity of 10 dB when it is 10 times more intense than I_2 , 20 dB when it is 100 times more intense, 30 dB when it is 1,000 times more intense, etc.

In this question, **Table 1** shows that the sound intensities for a 1.0 T MRI and a vacuum cleaner are 110 dB and 70 dB, respectively. Therefore, the intensity of the 1.0 T MRI is 40 dB higher than the vacuum cleaner:

$$110 \text{ dB} - 70 \text{ dB} = 40 \text{ dB}$$

A 40 dB increase in sound intensity corresponds to an increase by a factor of 10^4 . Therefore, the sound from a 1.0 T MRI is 10,000 times more intense than the sound from a vacuum cleaner.

(Choice A) The sound from a 1.0 T MRI is 40 dB more intense than the sound from a vacuum cleaner, not 40 times more intense.

(Choices B and C) A sound that is 40 dB louder is 10^4 times greater or 10,000 times more intense, not 100 or 1,000 times more intense.

Educational objective:

Sound intensity is typically measured in units of decibels (dB). The dB scale is logarithmic, so increasing the intensity of a sound by 10 dB corresponds to a factor of 10 increase in intensity.

Time spent: 0 sec

Subject:
Physics

QID: 403878

System:
Electrostatics and Circuits

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Equation 1

MRI can be combined with another imaging method, positron emission tomography (PET), to detect cancer. PET imaging maps the uptake of a radioactive metabolite. The radioactive decay emits a positron that annihilates with an electron and forms gamma rays that are imaged by the scanner.

How will the work done by the Lorentz force on an electron traveling through the magnetic field of an MRI scanner change if the magnetic field strength is increased?

- ✔ ☒ A. The work will not change, because it is always zero. [15%]
- ☐ B. The work will increase, because the velocity of the electron will increase. [15%]
- ☐ C. The work will increase, because the force on the electron will increase. [66%]
- ☐ D. The work will decrease, because the velocity of the electron will decrease. [3%]

Correct

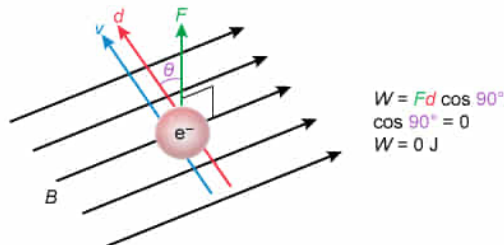
15% answered correctly

Explanation:

Work done by Lorentz force on electron

- W Work
- F Lorentz force
- d Displacement
- θ Angle between F and v
- v Speed of electron
- B Magnetic field

$$W = Fd \cos \theta$$



$$\begin{aligned} W &= Fd \cos 90^\circ \\ \cos 90^\circ &= 0 \\ W &= 0 \text{ J} \end{aligned}$$

$W = 0 \text{ J}$ because F is always perpendicular to d

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The Lorentz force F acts on a charge q moving in a magnetic field B with velocity v . The direction of F is perpendicular to both the direction of v and the direction of B . The magnitude of F equals the product of q , v , B , and the sine of the angle θ_B between the directions of v and B :

$$F = qvB \sin \theta_B$$

The work W done by F on an object equals the product of F , the magnitude of the object's displacement d , and the cosine of the angle θ between the directions of d and F :

$$W = Fd \cos \theta$$

In this question, the value of B increases, which causes F to increase. However, the value of θ is always 90° because d is the same direction as v , and F is always perpendicular to v . Therefore, W must always equal 0 J because $\cos 90^\circ$ is always equal to 0 .

(Choices B and D) Increasing the magnetic field strength does not change the velocity of the electron.

(Choice C) Increasing the magnetic field strength will increase the force on the electron, but zero work is done by the force because the force is perpendicular to the displacement of the electron.

Educational objective:

The work done by a constant force acting on an object equals the product of the force, the displacement of the object, and the cosine of the angle between the directions of the force and displacement. A force applied perpendicular to the direction of the displacement does zero work on the object.

Time spent: 0 sec
Subject:
Physics

QID: 403879
System:
Electrostatics and Circuits

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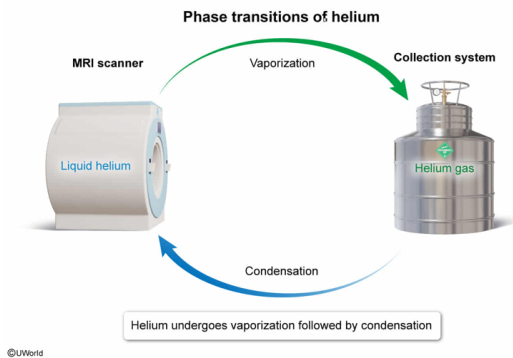
Which of the following describes the order of the phase changes when helium moves to the collection system and returns to the MRI scanner?

- ☐ A. Condensation followed by vaporization [7%]
- ✓ ☒ B. Vaporization followed by condensation [88%]
- ☐ C. Vaporization followed by deposition [2%]
- ☐ D. Deposition followed by vaporization [1%]

Correct

88% answered correctly

Explanation:



The **phase** of matter depends on its physical properties, such as its temperature or pressure. The three basic phases of matter are solid, liquid, and gas. The **phase transitions** refer to the two phases involved and the direction of the transition. The transition from a solid to a liquid is melting, and the reverse transition is freezing. Similarly, the transition from a solid to a gas is sublimation, and the reverse transition is deposition. Additionally, the transition from a liquid to a gas is vaporization, and the reverse transition is condensation.

In this question, helium boils out of the MRI scanner and then is cooled back into a liquid. Hence, helium undergoes **vaporization** from a liquid into a gas, followed by **condensation** from a gas back into a liquid.

(Choice A) The helium undergoes vaporization and is stored as a gas in the collection system. This is followed by condensation, when it is returned to the scanner as a liquid; it does not undergo condensation followed by vaporization.

(Choices C and D) The helium does not undergo deposition because it does not change phase from a gas into a solid.

Educational objective:

The three basic phases of matter are solid, liquid, and gas. The transition from liquid to gas is called vaporization, and the transition from gas to liquid is called condensation.

Time spent: 0 sec

Subject:
Physics

QID: 403880

System:
Electrostatics and Circuits

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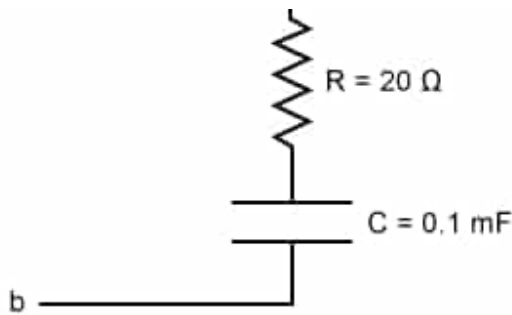
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A researcher wants to decrease the cutoff frequency of the RC filter circuit shown above by 50%. Would adding a 0.1 mF capacitor in series with the existing capacitor achieve the desired result?

- ☐ A. Yes, because the equivalent capacitance would be 0.05 mF. [24%]
- ☐ B. Yes, because the equivalent capacitance would be 0.2 mF. [17%]
- ✓ ☒ C. No, because the equivalent capacitance would be 0.05 mF. [42%]
- ☐ D. No, because the equivalent capacitance would be 0.2 mF. [15%]

Correct

41% answered correctly

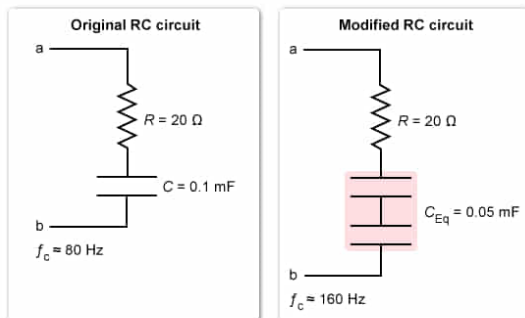
Explanation:

Cutoff frequency of an RC circuit

$$f_c = \frac{1}{2\pi RC_{Eq}}$$

$$\frac{1}{C_{Eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

- f_c Cutoff frequency
- R Resistance
- C_{Eq} Equivalent capacitance
- C_1 Original capacitor
- C_2 Added capacitor



f_c does not decrease by 50% because C_{Eq} is 0.05 mF

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Capacitors connected in **parallel** can be replaced with a single **equivalent capacitance** C_{Eq} by summing the individual capacitor values:

$$C_{Eq} = C_1 + C_2 + C_3$$

Similarly, capacitors connected in **series** can be replaced with C_{Eq} using the equation:

$$\frac{1}{C_{Eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$$

The **cutoff frequency** f_c for an RC filter circuit is calculated from the resistance R and capacitance C using Equation 1 from the passage:

$$f_c = \frac{1}{2\pi RC}$$

When the RC circuit contains more than one capacitor, f_c is calculated from C_{Eq} :

$$f_c = \frac{1}{2\pi RC_{Eq}}$$

In this question, $R = 20 \, \Omega$ and $C = 0.1 \, \text{mF}$ for the original RC filter circuit, yielding an f_c given by:

$$f_c = \frac{1}{2\pi(20)(0.1 \times 10^{-3})} = \frac{1000}{4\pi} \approx 80 \, \text{Hz}$$

If the researcher connected a second capacitor with $C = 0.1 \, \text{mF}$ in series with the original capacitor, this would yield a C_{Eq} of:

$$\frac{1}{C_{Eq}} = \frac{1}{0.1 \, \text{mF}} + \frac{1}{0.1 \, \text{mF}} = \frac{2}{0.1 \, \text{mF}}$$

$$C_{Eq} = \frac{0.1 \, \text{mF}}{2} = 0.05 \, \text{mF}$$

With $R = 20 \, \Omega$ and $C_{Eq} = 0.05 \, \text{mF}$, the f_c of the new RC filter circuit would be:

$$f_c = \frac{1}{2\pi(20)(0.05 \times 10^{-3})} = \frac{1000}{2\pi} \approx 160 \, \text{Hz}$$

Therefore, the researcher would not decrease the value of f_c by 50%, but would instead increase the f_c by a factor of two because the C_{Eq} of the new RC filter circuit would be 0.05 mF.

(Choices A and B) Connecting a 0.1 mF capacitor in series with the existing capacitor would not decrease the cutoff frequency by 50%; it would increase the cutoff frequency by a factor of two.

(Choice D) The equivalent capacitance of two 0.1 mF capacitors connected in series is 0.05 mF, not 0.2 mF.

Educational objective:

The equivalent capacitance of capacitors connected in parallel equals the sum of the individual capacitances. For capacitors connected in series, the inverse of the equivalent capacitance equals the sum of the inverses of the individual capacitances.

Time spent: 0 sec

QID: 403881

Subject:
Physics

System:
Electrostatics and Circuits

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According to the data in Table 1, which of the following makes noise at a frequency that resonates in the bore of an MRI magnet? (Note: The speed of sound in air is

350 m/s.)

- ✓ ☒ A. 1.0 T MRI [26%]
☐ B. 1.5 T MRI [30%]
☐ C. 3.0 T MRI [22%]
☐ D. Gas lawnmower [20%]

Correct

26% answered correctly

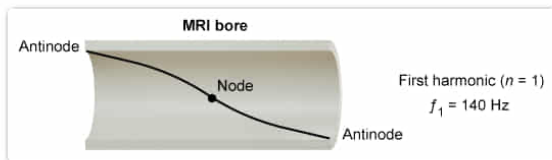
Explanation:

Resonance in MRI bore open on both ends

$$\lambda_n = \frac{2L}{n} \quad f_n = \frac{nv}{2L}$$
$$n = (1, 2, 3, \dots)$$

- λ_n Wavelength of n^{th} harmonic
- L Length
- f_n Frequency of n^{th} harmonic
- v Speed of sound

$$L = 1.25 \text{ m} \quad v = 350 \frac{\text{m}}{\text{s}} \quad f_n = \frac{n \left(350 \frac{\text{m}}{\text{s}} \right)}{2 (1.25 \text{ m})} = 140n \text{ Hz}$$



Sound from 1.0 T MRI resonates in bore of MRI scanner

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Standing sound waves form inside a pipe when the waves experience constructive interference and the sound resonates in the pipe. A pipe open on both ends **resonates** when the **wavelength** λ of sound is twice the length L of the pipe:

$$\lambda = 2L$$

This wavelength represents the **fundamental harmonic**, which is the longest resonant λ for the pipe. Additional harmonics are given by:

$$\lambda_n = \frac{2L}{n}$$

where the value of n is any positive integer, and λ_n is the n^{th} harmonic for the pipe.

The **frequency** f of a sound wave equals the ratio of the speed of sound v and λ :

$$f = \frac{v}{\lambda}$$

Therefore, the harmonic frequencies f_n for a pipe open on both ends are:

$$f_n = \frac{nv}{2L}$$

In this question, the L of the MRI magnet is 1.25 m and v equals 350 m/s, yielding:

$$f_n = \frac{n\left(350 \frac{\text{m}}{\text{s}}\right)}{2(1.25 \text{ m})} = 140n \text{ Hz}$$

Therefore, the fundamental resonant frequency f_1 equals 140 Hz, and any f that can be evenly divided by f_1 is a harmonic frequency of the magnet. From [Table 1](#), a 1.0 T MRI generates sound with an f of 700 Hz, which evenly divides into f_1 :

$$\frac{700 \text{ Hz}}{140 \text{ Hz}} = 5$$

Hence, the sound generated by the 1.0 T MRI equals the fifth harmonic of the MRI scanner.

(Choices B and C) The frequencies of these MRI scanners are not harmonics because 1,000 and 1,500 Hz are not multiples of the fundamental frequency, 140 Hz.

(Choice D) The frequency of a gas lawnmower is not a harmonic because 250 Hz is not a multiple of 140 Hz.

Educational objective:

The fundamental resonant frequency of a pipe open on both ends equals the ratio of the speed of sound and twice the length of the pipe. The higher resonant frequencies are all positive integer multiples of the fundamental resonant frequency.

Time spent:0 sec

Subject:
Physics

QID:403882

System:
Electrostatics and Circuits